

Using 3D Reconstruction to create Pervasive Augmented Reality Experiences: A comparison

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ABSTRACT

This paper presents a prototype for configuration and visualization of Pervasive Augmented Reality (AR) experiences using two versions: desktop and mobile. It makes use of 3D scans from physical environments to provide a reconstructed digital representation of such spaces to the desktop version and enable positional tracking for the mobile. While the desktop presents a non-immersive setting, the mobile provides continuous AR in the physical environment. Both versions can be used to place virtual content and ultimately configure an AR experience. The authoring capabilities of the proposed solution were compared by conducting a user study focused on evaluating their usability.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction Paradigms—Mixed/augmented reality;

Human-centered computing—Human computer interaction (HCI)—HCI Design and Evaluation Methods—User Studies

1 INTRODUCTION

Augmented Reality (AR) adds additional layers of digital information on top of real environments [1]. Pervasive AR extends this concept through experiences that are continuous in space, being aware of and responsive to the user's context and pose [2, 3]. It has the potential to help AR evolve from an application with a unique purpose to a multi-purpose continuous experience that changes the way users interact with information and their surroundings.

Nevertheless, these type of experiences pose several challenges [2, 3], namely **tracking of the user position**, i.e., understanding the user location in relation to the environment, **content management**, i.e., making available different types of digital content (e.g., text, images, videos, 3D models, sound), **adequate configuration of virtual content**, i.e., configuring experiences by choosing where augmented content is placed in the real-world, **uninterrupted display of information**, i.e., making available these continuous experiences, in order for them to be explored by the target audiences through adequate mechanisms to display, filter and select the relevant information at a given moment, as well as use of **suitable interaction**, i.e., providing suitable interaction techniques according to the devices being used, which allow easy-to-use authoring features.

In this paper, we propose a solution to create and explore engaging Pervasive AR experiences for indoor environments. We take advantage of a 3D scan from a physical space that can be configured in a desktop or mobile setting according to the requirements of the scenario being addressed. Both versions of the prototype were evaluated through a user study, based on usability and usefulness.

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2 PERVASIVE AUGMENTED REALITY PROTOTYPE

The proposed prototype consists of various components that work towards creating Pervasive AR experiences (Figure 1). It was built using the Unity game engine, integrated with the Vuforia Library. First, it is necessary to select the environment that will be augmented. Then, using a 3D scanner, a reconstruction is obtained, i.e., a virtual replica or Area Target (AT) (a data set containing information associated with tracking, 3D geometry, occlusion and collision, which can be imported into Unity) (Figure 1 - 5). Afterwards, the AT is uploaded to Vuforia, obtaining an improved 3D model with potential for clearer use in Unity, designated as Vuforia Area Target (VAT). On top of the VAT, pre-defined virtual models can be selected and manipulated to provide additional layers of information, relevant to the use-case being addressed. This process may be conducted with different interaction methods: desktop or mobile setting. Finally, the experience is ready to be explored in the real-world environment.



Figure 1: Creation process of Pervasive AR experiences: 1- real-world environment; 2- reconstruction of the physical space; 3- integration of pre-existing augmented content according to the experience scope; 4- visualization of continuous AR-based content in the physical environment; 5- overview of the steps need to generate the experience.

For manipulation of virtual content during configuration, two methods exist. The **desktop** version provides a computer interface, taking advantage of the VAT to position objects in relation to the digital model. The navigation within this model simulates a user walking in the corresponding physical location. The viewpoint is placed inside the VAT model and traverses it at a fixed height from the perceived ground, following gravity and using the collision mesh to replicate a real interaction with the environment. Only one virtual model may be selected and controlled at a time. The object follows the viewing point controlled by mouse movement. This is done by casting a ray through the center of the screen onto the mesh of the virtual scene and placing that object onto the first plane the ray cast intersects (Figure 2 - 1). As the user moves the camera viewpoint, the object follows that movement by being continuously projected onto the planes of the scene, as if sliding over the surfaces of that environment. Along with the translation, the user is also able to control the object's rotation by scrolling the mouse wheel. This AR-based experience can be previewed in the simulated scenario of this setting, and experienced in the physical space with the mobile version. The **mobile** version uses the VAT to

track the device location and present the content accordingly. The navigation is done as the user moves through the physical space and controls the camera's viewpoint. As the device's camera captures the environment, this information is compared to the VAT data, enabling recognition of geometry and textures. The user also controls a single object at a time. A ray is cast through the center of the screen, perpendicular to the camera, placing the object upright on the first surface hit, according to its normal vector and using the knowledge and recognition of the VAT geometry. In addition, the object's rotation is tied to the device's tilt. As the device rotates around the axis perpendicular to its camera, so does the manipulated object around the axis corresponding to the normal vector of the surface it is onto (Figure 2 - 2).

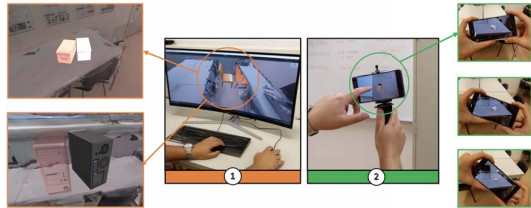


Figure 2: Virtual content configuration: Left (1) - VAT model using a desktop computer; Right (2) - physical space using a handheld device.

3 USER STUDY AND RESULTS

A user study was conducted to evaluate the usability, focusing on user navigation and content placement using conditions: C1- desktop and C2- mobile. The goal was to position 6 virtual objects in the surrounding space (Figure 3). Task performance was collected: time needed to complete all procedures, logged in seconds by the devices used; differences in position and rotation in relation to the corresponding target pose, calculated and registered by the devices. Participants' opinion was gathered through post-task questionnaire and interview, including: demographic information, SUS, the NASA-TLX, user preferences regarding each condition's usefulness and ease of use. Participants were instructed on the experimental setup, the tasks, and gave their informed consent. They were introduced to the prototype and a time for adaptation was provided. All participants performed the task using both conditions in different orders to minimize learning effects. At the end, participants answered the post-task questionnaire and a small interview was conducted to understand their opinion towards the conditions used.

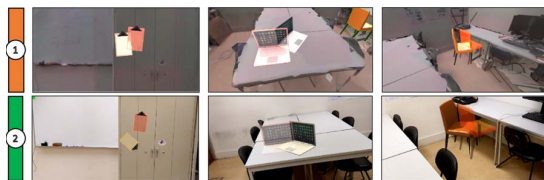


Figure 3: Positioning of 3 virtual objects and their respective target poses (transparent counterpart): Top (1) - content configuration on the 3D reconstruction using the desktop version; Bottom (2) - content configuration on the physical space through the mobile version.

We recruited 17 participants (3 female) having previous experience with evaluating and using AR-based applications. On average, tests lasted 17 minutes, including the aforementioned steps (tasks took 8 minutes to complete). Overall, participants were quicker and more precise with the desktop condition. The placement task was 2.77 times faster, 4.12 times more precise in translation and 10.08 times in rotation with the desktop condition (on average). For larger

models, such as the chair (Figure 3) the mobile condition proved to be harder to place accurately, likely due to requiring an increased distance from the device to the desired location. The SUS revealed a score of 92 for the desktop and 76 for the mobile condition, implying excellent and good usability, respectively. Additionally, the NASA-TLX indicated that the desktop condition had a lower mental and physical demand, while the mobile had more average demands. Nevertheless, both methods were agreed to have potential for improved performance with training or further experience. For the **mobile condition**, although some users found the object rotation approach intuitive, most struggled with understanding the necessary movement, suggesting that further research in alternative methods may be required. Yet, this condition will likely provide users with a better perception of the final product, since they will essentially be experiencing it during the creation process. The **desktop condition** results may be caused by participants' greater experience with computer interfaces. Since the number of errors was almost 0 in the placement of objects for either method, it can also be argued that both platforms are reliable and satisfactory authoring tools according to the use-case. Moreover, this version may be used to configure remote environments, avoiding the need to be present at the desired location, especially relevant for dynamic environments with multiple users, vehicles, or robots (e.g., logistics warehouse, assembly lines, etc.). Besides the useful remoteness aspect, it could also prove advantageous for handling the creation of larger, or simply more, experiences in the same time, with less physical effort. Thus, it is thought to have better scalability. Perhaps, a **combination of conditions** would be a good approach. The experience would be created through the mobile method, and then the desktop one could be employed to update the experience with new models and make tweaks. This version could also be extended to immersive experiences in VR, allowing direct manipulation using 6DOF controllers.

4 CONCLUDING REMARKS AND FUTURE WORK

We proposed a prototype for configuration and visualization of Pervasive AR experiences. The desktop version excels in precision and practicality benefiting from a 3D reconstruction of the physical space and a familiar interface, providing accurate control and manipulation of virtual content. The mobile version provides a process closer to the final experience, granting better perception on how the augmented content will be envisioned. Hence, both versions appear as a valid solution (according to the scenario and requirements being addressed) towards creating pervasive AR-based experiences, which users can configure, explore, interact with and learn from.

This study is being expanded by improving the authoring features, which can result in more complex and interactive AR experiences. We will also move into a dynamic industrial scenario, which presents a richer environment with additional challenges, like handling operators movement, as well as re-positioning of physical objects.

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